

DSC8011: SOCIAL FOUNDATIONS OF DATA SCIENCE

Effective Term

Semester A 2026/27

Part I Course Overview

Course Title

Social Foundations of Data Science

Subject Code

DSC - Data Science

Course Number

8011

Academic Unit

Data Science (DS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

R8 - Research Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

SDSC8011 Social Foundations of Data Science

Exclusive Courses

Nil

Part II Course Details

Abstract

As a large part of the current data used by data scientists are created by or about humans, it is necessary and helpful to introduce to the students of data science social processes and mechanisms underlying human data. In particular, the course focuses on key issues that give rise to various challenges and/or biases in human data such as self selection, social desirability, cognitive limitations, nonindependence, contextual effects, ecological fallacy, and etc. The disposition of these issues will be inherently linked to technical issues such as sampling, measure errors, statistical control, causal inference, and etc. At the end of the course, the students are expected to be able to identify potential human errors or biases in the existing data science literature, develop systematic approaches to addressing the problems, and implement some of the suggested remedies with real world data.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Explain clearly fundamental characteristics of the processes and mechanisms underlying the creation, analysis, and interpretation of human data	10	x		
2	Discuss existing data science literature that involves human data to identify common symptoms, causes and consequences of human data with fatal social and/or cognitive biases	20	x	x	
3	Design appropriate procedures, measures, analytical approaches, and interpretation rules to address common problems in human data	30	x	x	x
4	Apply the proposed remedies to real world applications, demonstrating the effectiveness and resulting.	40	x	x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture	Students will engage in formal lectures and relevant class discussions.	1, 2, 3, 4	39 hours in total
2	Critical review of relevant literature	Students will engage in learning through critiques of common problems in the data science literature involving human data.	1, 2, 3, 4	in or after classes

3	Research design assignments	Students will work on in-class or take-home assignments involving the development and design of remedy procedures, measure, analytical approaches, and interpretation rules.	1, 2, 3, 4	in or after class
4	Empirical research paper	Students will engage in actual implementations of the proposed remedies and writing up the results in a publishable quality.	1, 2, 3, 4	after class

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- for nil entry)	Allow Use of GenAI?
1	Critique Essays Students are required to review a variety of published studies that involve potential human errors with sampling, measurement, analysis, or interpretations.	1, 2	20	-	Yes
2	Design Reports Students are required to develop specific remedies to address the identified common problems in the literature.	1, 2, 3	30	-	Yes
3	Research Paper Students are required to interpret the proposed remedies in real world application contexts.	1, 2, 3, 4	50	-	Yes

Continuous Assessment (%)

100

Examination (%)

0

Assessment Rubrics (AR)**Assessment Task**

Critique Reviews (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to identify the symptoms, causes, and consequences of various errors and biases in human data.

Excellent

A+, A, A-

Good

B+, B, B-

Fair

C+, C, C-

Marginal

D

Failure

F

Assessment Task

Design Assignments (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to propose appropriate procedures, instruments, and test measures to address common errors and biases in human data.

Excellent

A+, A, A-

Good

B+, B, B-

Fair

C+, C, C-

Marginal

D

Failure

F

Assessment Task

Research Paper (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to carry out one or more empirical study(ies), implementing the proposed remedies to demonstrate improved quality in the resulting findings.

Excellent

A+, A, A-

Good

B+, B, B-

Fair

C+, C, C-

Marginal

D

Failure

F

Assessment Task

Critique Reviews (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to identify the symptoms, causes, and consequences of various errors and biases in human data.

Excellent

A+, A, A-

Good

B+, B

Marginal

B-, C+, C

Failure

F

Assessment Task

Design Assignments (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to propose appropriate procedures, instruments, and test measures to address common errors and biases in human data.

Excellent

A+, A, A-

Good

B+, B

Marginal

B-, C+, C

Failure

F

Assessment Task

Research Paper (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to carry out one or more empirical study(ies), implementing the proposed remedies to demonstrate improved quality in the resulting findings.

Excellent

A+, A, A-

Good

B+, B

Marginal

B-, C+, C

Failure

F

Part III Other Information

Keyword Syllabus

Characteristics of human data, self-selection biases, social desirability biases, cognitive biases, aggregation biases, found data, made data, physical control vs. statistical control, causal inference, sampling of social units, research design, online experiment, behaviour analytics, text mining, research ethics.

Reading List

Compulsory Readings

Title	
1	Babbie, E. (2007). The practice of social research, 11th edition. Wadsworth Publishing. Reprint, Tsinghua University Press.
2	Salganik, M. J. (2018). Bit by bit: Social research in the digital age. Princeton University Press.

Additional Readings

Title	
1	Lazer, D., Pentland, A. S., Adamic, L., Aral, S., Barabasi, A. L., Brewer, D., ... & Jebara, T. (2009). Life in the network: the coming age of computational social science. <i>Science</i> (New York, NY), 323(5915), 721.
2	Watts, D. J. (2013). Computational social science: Exciting progress and future directions. <i>The Bridge on Frontiers of Engineering</i> , 43(4), 5-10.
3	Blumenstock, J., Cadamuro, G., & On, R. (2015). Predicting poverty and wealth from mobile phone metadata. <i>Science</i> , 350(6264), 1073-1076.
4	Bond, R. M., Fariss, C. J., Jones, J. J., Kramer, A. D., Marlow, C., Settle, J. E., & Fowler, J. H. (2012). A 61-million-person experiment in social influence and political mobilization. <i>Nature</i> , 489(7415), 295.
5	Kwak, H., Lee, C., Park, H., & Moon, S. (2010, April). What is Twitter, a social network or a news media?. In <i>Proceedings of the 19th international conference on World wide web</i> (pp. 591-600). ACM.
6	Ortiz-Ospina, E., & Roser, M. (2018). Happiness and life satisfaction. <i>Our World in Data</i> [online resource].
7	Salganik, M. J., Dodds, P. S., & Watts, D. J. (2006). Experimental study of inequality and unpredictability in an artificial cultural market. <i>Science</i> , 311(5762), 854-856.
8	Wu, S., Hofman, J. M., Mason, W. A., & Watts, D. J. (2011, March). Who says what to whom on twitter. In <i>Proceedings of the 20th international conference on World wide web</i> (pp. 705-714). ACM.

9	Kramer, A. D., Guillory, J. E., & Hancock, J. T. (2014). Experimental evidence of massive-scale emotional contagion through social networks. <i>Proceedings of the National Academy of Sciences</i> , 201320040.
10	Salganik, M. J., Dodds, P. S., & Watts, D. J. (2006). Experimental study of inequality and unpredictability in an artificial cultural market. <i>Science</i> , 311(5762), 854-856.
11	Zhu, J. H., Milavsky, J. R., & Biswas, R. (1994). Do televised debates affect image perception more than issue knowledge? A study of the first 1992 presidential debate. <i>Human Communication Research</i> , 20(3), 302-333.
12	Reinemann, C., & Maurer, M. (2005). Unifying or polarizing? Short-term effects and postdebate consequences of different rhetorical strategies in televised debates. <i>Journal of Communication</i> , 55(4), 775-794. (newly added)
13	Liang, H., & Zhu, J. J. H. (2017). Big data, collection of (social media, harvesting). In J. Matthes, C. S. Davis, & R. F. Potter (Eds.), <i>International Handbook of Communication Methods</i> , Wiley & Sons.
14	Leskovec, J., & Faloutsos, C. (2006, August). Sampling from large graphs. In <i>Proceedings of the 12th ACM SIGKDD international conference on Knowledge discovery and data mining</i> (pp. 631-636). ACM. (newly added)
15	Morstatter, F., Pfeffer, J., Liu, H., & Carley, K. M. (2013, July). Is the sample good enough? Comparing data from Twitter's streaming API with Twitter's firehose. In <i>ICWSM</i> . (newly added)
16	Xu, X. K., & Zhu, J. J. H. (2016). Flexible sampling large-scale social networks by self-adjustable random walk. <i>Physica A: Statistical Mechanics and its Applications</i> , 463, 356-365.
17	Zhu, J. J. H., Mo, Q., Wang, F., & Lu, H. (2011). A random digit search (RDS) method for sampling of blogs and other web content. <i>Social Science Computer Review</i> , 29(3), 327-339.
18	Ginsberg, J., Mohebbi, M. H., Patel, R. S., Brammer, L., Smolinski, M. S., & Brilliant, L. (2009). Detecting influenza epidemics using search engine query data. <i>Nature</i> , 457, 1012-U1014.
19	Lazer, D., Kennedy, R., King, G., & Vespignani, A. (2014). The Parable of Google Flu: Traps in Big Data Analysis. <i>Science</i> , 343, 1203-1205.
20	Burke, M., & Kraut, R. E. (2014). Growing closer on Facebook: changes in tie strength through social network site use. <i>Proceedings of the 32nd annual ACM conference on Human factors in computing systems</i> , Toronto, Ontario, Canada. http://thoughtcrumbs.com/publications/burke_chi2014.pdf .
21	Xu, P. P., Wu, Y. C., Wei, E. X., Peng, T. Q., Liu, S. X., Zhu, J. J. H., & Qu, H. M. (2013). Visual Analysis of Topic Competition on Social Media. <i>IEEE Transactions on Visualization and Computer Graphics</i> , 19(12), 2012-2021.
22	Aral, S., & Nicolaides, C. (2017). Exercise contagion in a global social network. <i>Nature Communications</i> , 8, 14753.
23	Klein, K. J., Dansereau, F., & Hall, R. J. (1994). Levels issues in theory development, data collection, and analysis. <i>Academy of Management review</i> , 19(2), 195-229.
24	Ackland, R., & Zhu, J. J. H. (2015). Social network analysis. In P. Halfpenny & R. Procter (Eds.), <i>Innovations in digital research methods</i> (pp. 221-244). Sage Publications.
25	Murdock, J., Allen, C., & DeDeo, S. (2017). Exploration and exploitation of Victorian science in Darwin's reading notebooks. <i>Cognition</i> , 159, 117-126.
26	Peng, T. Q., Liu, M. C., Wu, Y. C., & Liu, S. X. (2016). Follower-followee Network, Communication Networks and Vote Agreement of U.S. Members of Congress. <i>Communication Research</i> , 43, 996-1024.
27	Zhang, L., Zheng, L., & Peng, T. Q. (2017). Structurally embedded news consumption on mobile news applications. <i>Information Processing & Management</i> , 53, 1242-1253.
28	Zhu, J. J. H., Chen, H. X., Peng, T. Q., Liu, X. F., & Dai, H. X. (2018). How to measure sessions of mobile device use? Quantification, Evaluation, and Applications. <i>Mobile Media & Communication</i> , 6, 215-232.
29	boyd, danah (2016). Untangling research and practice: What Facebook's emotional contagion study teaches us. <i>Research Ethics</i> , 12, 4-13.
30	Goroff, D. L. (2015). Balancing privacy versus accuracy in research protocols. <i>Science</i> , 347, 479-480.
31	Open Science Collaboration. (2015). Estimating the reproducibility of psychological science, <i>Science</i> , 349, aac4716.